

Analysis of the Impacts Observed During the Festival Cruïlla Demos

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Contents

1	Discussion	2
1.1	Hypersexualization of Women	2
1.2	'Adultification' of Children and Adolescents	2
1.3	Racial Stereotyping and Ethnic Distortion	3
1.4	Alteration of Gender Identity	4
2	Conclusions	4

1 Discussion

Advances in artificial intelligence have led to its widespread and democratised use across a heterogeneous population with diverse sensitivities, both in professional and leisure contexts. Yet despite this evolution, AI remains far from perfect, and within this ongoing process of refinement, its occasional lack of rigour may give rise to conflicts within the emerging human–AI interaction (HAI). The advent of generative AI technologies has introduced a range of ethical challenges: firstly, because they tend to reason in an overly simplistic manner when confronted with complex realities; and secondly, because they learn from data drawn from past situations, leading AI systems to make statistical inferences [16]. Addressing the risks and harms that arise from these limitations is crucial—not only because they have already caused measurable impact and are likely to persist, but also because they may intensify and accumulate over time [13].

The AI Risk Management Framework (AI RMF), developed by the United States National Institute of Standards and Technology (NIST), emphasises the *identification* and *prioritisation* of harms as critical steps in managing AI-related risks [17]. Although a growing body of research seeks to identify and categorise risks and harms across different domains and applications—mapping them within various taxonomies [3, 4, 5, 8, 14, 20, 21, 25, 27]—much of this work still lacks grounding in empirical evidence and rarely addresses *who* is affected or *how* risk and harm emerge, thereby limiting its value for risk prioritisation and mitigation [13]. While earlier studies have proposed frameworks to identify points of technological failure [6, 12, 15, 18, 22], most focus narrowly on technical aspects, lack an empirical basis, and overlook how generative AI introduces a distinct landscape of failures and risks compared with other forms of artificial intelligence [13].

Drawing on observations from the prototypes deployed in the Cruïlla Festival environment, this study identifies and analyses several instances of *representational harms*—cases in which systems reinforce stereotypes or render specific groups invisible [23], among other impacts. These were grouped into four subcategories according to their area of impact: hypersexualisation of women, ‘adultification’ of children and

adolescents, ethnic distortion and racial stereotyping, and alteration of gender identity. Based on these findings, the paper aims to elucidate the challenges posed by the emergence of this technology, highlight the ethical considerations necessary for its responsible use, and suggest possible pathways for improvement.

1.1 Hypersexualization of Women

The hypersexualisation of women emerged as one of the most evident and consequential effects observed during the fieldwork at the festival. This phenomenon, detected exclusively among female participants and in the images produced by the CVC prototype, manifested through the exaggeration of certain body features and, in some instances, the partial removal of clothing—both factors that further amplified the visual impact. Coen Antens, a representative of the CVC technical team at Cruïlla, explained that the ethical issue was identified when they ‘noticed that the AI was enlarging certain body parts, particularly in cases involving lighter clothing’ [1].

Although these distortions were clearly visible to external observers, many participants failed to perceive the physical alterations made to their own bodies. Age proved to be a decisive factor in recognizing hypersexualisation: older participants tended to detect and react to it more rapidly and negatively, whereas most younger participants expressed little or no discomfort with the generated images (Figure 1).

The findings thus point to a significant risk in the evolution and democratisation of artificial intelligence, stemming from users’ inability to recognise distortions in their own image. When projected onto human bodies, such distortions can foster discrimination, self-rejection, and the erasure of any form that diverges from the aesthetic ideals perpetuated by AI.

1.2 ‘Adultification’ of Children and Adolescents

When examining the child demographic—one of the groups most affected by the prototypes displayed at the festival—another form of harm becomes evident, this time caused by both prototypes, although most pronounced in the case of the CVC model. In this instance, the generative AI exhibited a clear process of ‘adultification’, portraying all child participants in the same manner as adults. This not only involved the phenomenon of ‘adultification’ itself, but also the hypersexualisation



Figure 1: Perceived fidelity of the prototypes' outputs among women, in relation to the presence of visual biases, broken down by age.

Source: own elaboration based on data from the user survey.

of participating girls, similar to that observed among adult women.

This 'adultification' affected all children, as the model had not been calibrated for younger profiles [1]. Yet, as in previous cases, neither the underage participants nor their accompanying adults appeared to recognise this bias. In some cases, even when warned beforehand about the potential issue, adults dismissed the impact as merely 'fun and curious'. Responses collected from interviews with project participants further indicate that this age group perceived a high level of fidelity in the generated results—interpreting the images as accurate and failing to identify any bias.

Once again, the findings from the Cruilla Festival regarding the distorted reproduction of younger participants highlight a considerable risk for those

in late childhood and early adolescence. Throughout life, body image is in permanent (de)construction [11]; however, it is during childhood that concerns about weight, body-related beliefs, and appearance-focused behaviours often begin to emerge [24, 9]. Representations such as those produced by generative AI—and, in this case, by the Chair's prototypes—may negatively shape self-perception, as they not only 'adultify' but also impose a canonical beauty standard across all body types, distorting the diversity of the input reality and producing an alarmingly homogeneous representation.

1.3 Racial Stereotyping and Ethnic Distortion

When discussing racial bias in generative AI, reference is often made to cases in which the technology privileges certain ethnic groups over others—typically prioritising

the representation of the Western white male [10]—or reinforces racial and gender stereotypes when portraying professions [7, 26, 19], among other examples.

In the context of the Cruïlla Festival, this form of racial stereotyping manifested through the exaggeration of facial features, clothing styles, and other aesthetic elements that stereotyped the ethnicity attributed to each participant. On this occasion, the effect originated from the UAB students’ prototype, which, when identifying certain physical traits or specific preferences in the preliminary survey, assigned a series of ‘typical’ attributes associated with a given ethnic group.

Although such racial stereotyping was present, it did not constitute one of the most pronounced impacts observed across the prototypes—either in frequency or in intensity—as evidenced by participants’ responses regarding the perceived accuracy of the generated images. In this case, no bias was recognized by the users (Figure 2).

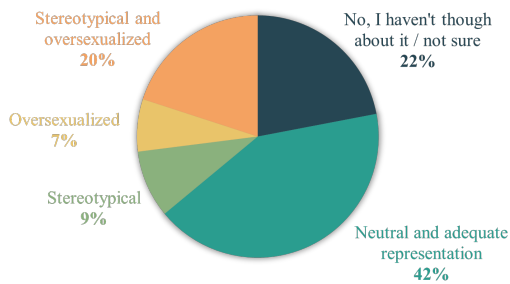


Figure 2: Perceived fidelity of the prototypes’ outputs in relation to the presence of visual biases.

Source: own elaboration based on data from the user survey.

In addition to the effects of racial stereotyping, the prototype also exhibited instances of ethnic distortion, particularly during its early iterations. In this case, the issue primarily involved alterations to skin tone and facial features, resulting in an inaccurate ethnic representation. During the first day of the festival, however, the prototype’s technical team identified that the error was likely human in origin: inadequate calibration of lighting during photo capture had caused the system to interpret skin tones as darker than they were, consequently generating representations aligned with African-American ethnicity. Another factor observed during the festival that contributed to the same distortion was dark, curly hair, which the AI similarly associ-

ated with that ethnic group.

A minor adjustment to the lighting setup considerably improved the results, although it did not prevent distortions towards other ethnic groups, such as Asian profiles—a detail that some participants found unsettling.

1.4 Alteration of Gender Identity

The final impact identified during the festival was the alteration of gender identity, once again originating from the UAB prototype. Although this form of representational harm occurred less frequently, it appears to have produced the strongest reactions among those affected. As with the case of hypersexualisation, the gender alteration exclusively affected female participants. The most noticeable change involved the emergence of facial hair (moustache or beard), typically accompanied by a masculinisation of facial features.

2 Conclusions

Over the three days of the Cruïlla Festival, both prototypes attracted a substantial number of participants. Alongside the survey already discussed, several informal conversations were held with attendees as part of the action research methodology underpinning this study. These exchanges provided the basis for the preliminary assessments and analyses from which key insights were drawn.

It became evident that generative artificial intelligence serves as a mirror of the values upheld by contemporary society. As outlined earlier, AI systems are trained on data derived from past contexts, leading them to replicate both gender and ethnic biases in their outputs. In the case of our prototypes, a pronounced impact was observed among female participants: hypersexualisation was present in all cases, and in some instances, even the alteration of gender identity—forms of representational harm that affected women exclusively (as illustrated by the gender-disaggregated perception of fidelity, Figure 3).

In certain cases, this impact deeply affected participants on a personal level. One respondent aged over 55 expressed shock at the ‘botox and breasts’ she perceived, voicing concern and disapproval ‘for the kids’ [2]. Another participant, aged between 45 and 54, stated that she did not recognise herself in the generated image,

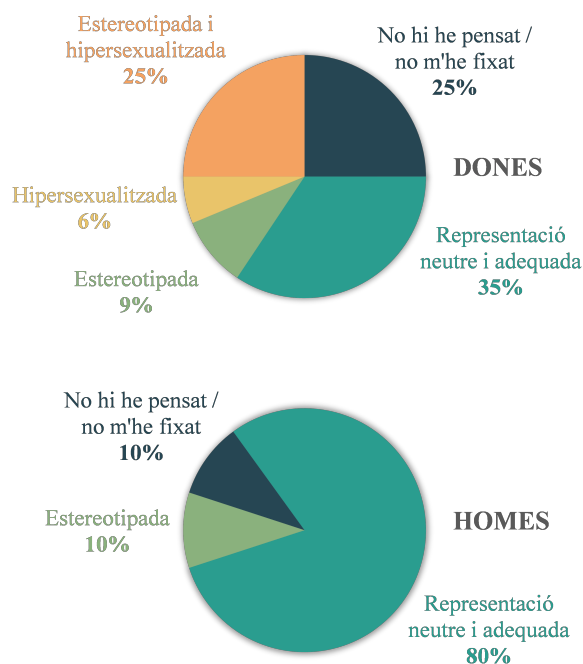


Figure 3: Perceived fidelity of the prototypes' outputs in relation to the presence of visual biases, broken down by gender.

Source: own elaboration based on data from the user survey.

commenting in an *in situ* conversation: 'This body has never been mine; I do not feel represented in the image. It is remarkable that such advanced technology should sexualise women's bodies in this way. My husband fell in love with me as I am, and we do not like the person shown in that image. We must fight to value all bodies and to resist these trends' [2].

Despite the initial discomfort, it is critical to communicate current understandings of the capabilities, limitations, and harms of Generative AI in ways that are comprehensible to the general public. [13]. Doing so allows emotionally difficult situations to be transformed into opportunities for collective learning—one of the central aims of this Chair. As Coen Antens observed during his interview at the festival, 'participating in the Chair has made us more rigorous: we explain to people that AI has biases, and it is our responsibility to make that evident' [1]. Similarly, Carlota Criado, a member of the UAB student team, described the mixed reactions of participants: 'you either take it as something amusing, or you struggle with it, saying "I am not this person", and if I am, I do not like myself' [1].

This duality in the impact of representational harms

has also emerged as a particularly interesting analytical element. A second finding from this study is that the vast majority of participants—regardless of whether they recognised the presence of bias—rated the prototypes very positively, with an average satisfaction score of 4.39 out of 5 (Figure 4), rising to 4.5/5 among those who tested both prototypes (Figure 5).

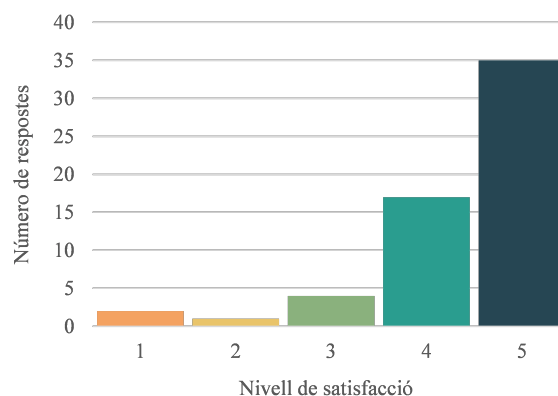


Figure 4: Distribution of satisfaction levels among respondents (1–5 scale).

Source: own elaboration based on data from the user survey.

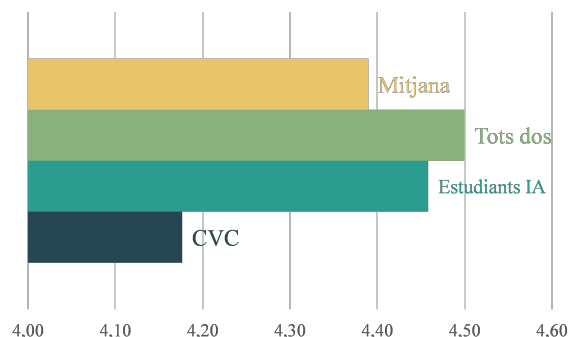


Figure 5: Distribution of satisfaction levels among participants (1–5 scale), by prototype.

Source: own elaboration based on data from the user survey.

Such high satisfaction levels suggest that participants attributed little importance to the biases present in the final outputs—a phenomenon that can be regarded as potentially dangerous given its broader implications. It is essential to remain critically aware of the risks that AI currently poses, not only to identify them, but also to address and mitigate them, particularly considering their potential long-term impact on future generations, who will never have known a world without artificial intelligence.

Finally, it is important to emphasise the value of spaces such as the Cruïlla Festival: ephemeral cities functioning as open laboratories where, through educational engagement, the public can gain exposure not only to cutting-edge technologies but also to the risks they entail. As reported by most respondents, their knowledge of Generative AI increased after visiting the buzz (Figure 6)—a highly positive outcome given the limited interaction time required by the prototypes.

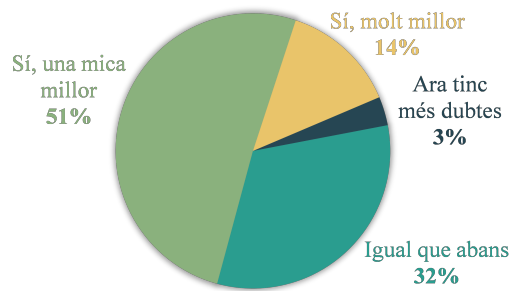


Figure 6: Participants' knowledge of Generative AI after engaging with the UAB buzz.

Source: own elaboration based on data from the user survey.

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